

- 3 (a) Fig. 3.1 is a diagram of an area of phloem tissue from a transverse section through the stem of a squash plant, *Cucurbita pepo*.

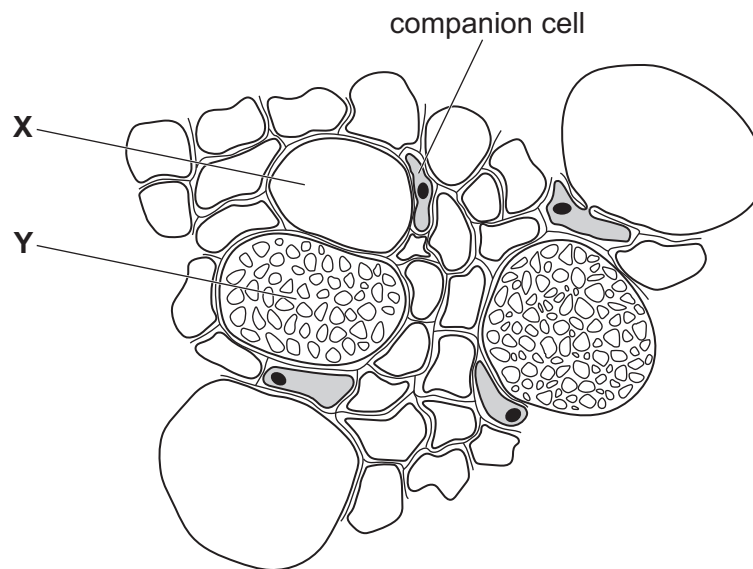


Fig. 3.1

- (i) Cell **X** and cell **Y** in Fig. 3.1 are sieve tube elements.

Explain why cell **X** and cell **Y** have very different appearances in this transverse section.

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- (ii) Sucrose is formed from the glucose synthesised by mesophyll cells in the leaves of *C. pepo*.

Explain how companion cells are involved in the transfer of sucrose into phloem sieve tubes.

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- (b) Hydrogen bonding is important in the movement of water in xylem.

- (i) Explain how hydrogen bonding occurs between two water molecules.

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- (ii) Outline how hydrogen bonding is involved in water transport in the xylem of a plant stem.

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- (iii) Hydrogen bonding between water molecules gives water a relatively high latent heat of vaporisation.

Suggest why it is important to plants that water has a high latent heat of vaporisation.

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[Total: 13]

Question 4 starts on page 12.